Carolyn A. Raithel

Curriculum Vitae
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EDUCATION

PhD Candidate in Astronomy and Astrophysics
Department of Astronomy, University of Arizona, Tucson, AZ
Expected Graduation Date: Summer 2020
Cumulative GPA: 3.88 on a 4.0 scale

Bachelor of Arts, Physics, June 2015
Carleton College, Northfield, Minnesota
Graduated Magna Cum Laude
Cumulative GPA: 3.92 on a 4.0 scale
Physics GPA: 3.97 on a 4.0 scale

AWARDS AND HONORS

NSF Graduate Research Fellow, 2016 – Present
P.E.O. Scholar Award Recipient, 2018
University of Arizona College of Science Outstanding Research Award, 2018
University of Arizona Department of Astronomy Outstanding Research Award, 2018
University of Arizona Roy P. Drachman Galileo Circle Scholarship, 2018
Theoretical Astrophysics Program Student Travel Grant, 2017
International Astronomy Union Travel Grant, 2017
American Physical Society Leroy Apker Award Finalist, 2015
Phi Beta Kappa Member, 2015
Sigma Xi Member, 2015
Graduated from Carleton College Magna Cum Laude, 2015
Distinction in the Carleton Physics Senior Integrative Exercise, 2015
Distinction in the Carleton Physics Major, 2015
NSF Graduate Research Fellowship Program Honorable Mention, 2015
Goldwater Scholar, 2014
Carleton College Dean's List, 2011-2012, 2013-2014
Distinction on the Carleton College Writing Portfolio, 2013

PUBLICATIONS

C. A. Raithel, "Constraints on the Neutron Star Equation of State from GW170817," invited review for the EPJA Topical Issue on the first Neutron Star Merger Observation - Implications for Nuclear Physics, 2019. In Press.

C. A. Raithel, F. Özel, and D. Psaltis. "Finite-Temperature Extension for Cold Neutron Star Equations of State," The Astrophysical Journal, Vol. 875, Issue 1, article id. 12 (2019).

C. A. Raithel, F. Özel, and D. Psaltis. "Tidal Deformability from GW170817 as a Direct Probe of the Neutron Star Radius," The Astrophysical Journal Letters, Volume 857, Issue 2, article id. L23 (2018).

C. A. Raithel, T. Sukhbold, and F. Özel. "Confronting Models of Massive Star Evolution and Explosions with Remnant Mass Measurements," The Astrophysical Journal, Volume 856, Issue 1, article id. 35 (2018).

C. A. Raithel, F. Özel, and D. Psaltis. "From Neutron Star Observables to the Equation of State. II. Bayesian Inference of Equation of State Parameters," The Astrophysical Journal, Vol. 844, Issue 2, 156 (2017).

C. A. Raithel, F. Özel, and D. Psaltis. "From Neutron Star Observables to the Equation of State. I. An Optimal Parametrization," The Astrophysical Journal, Vol. 831, Issue 1, 44 (2016).

C. A. Raithel, F. Özel, and D. Psaltis. "Model-Independent Inference of Neutron Star Radii from Moment of Inertia Measurements," Physical Review C, Vol. 93, Issue 3, id.032801 (2016).

C. A. Raithel, R. M. Shannon, S. Johnston, and M. Kerr. "Two Radio Emission Mechanisms in PSR J0901-4624," The Astrophysical Journal Letters, Vol. 804, Issue 1, L18 (2015).

R. M. Shannon et al. [incl. Raithel]. "Limitations in timing precision due to single-pulse shape variability in millisecond pulsars," MNRAS, Vol. 443, Issue 2, p.1463-1481 (2014).

J. M. Weisberg, N. Bamberger, C. A. Raithel, and Y. Huang. "Mapping the Emission Beam of Binary Pulsar B1913+16 via Geodetic Spin Precession," in preparation to submit to The Astrophysical Journal.

INVITED PRESENTATIONS

"Constraints on the Neutron Star Equation of State from GW170817." APS April Meeting, Denver, CO. April 2019.

"Constraints on the Neutron Star Equation of State from GW170817." Center for Nuclear Research Seminar, Kent State University, OH. January 2019.

"Journey to the Center of a Neutron Star: From Astrophysical Observations to the Equation of State." Plenary talk at the Ohio Section APS meeting, Toledo, OH. September 2018.

CONTRIBUTED PRESENTATIONS

"Finite-Temperature Equation of State for use in Neutron Stars Merger Simulations." Talk at COSPAR Assembly in Pasadena, CA. July 2018.

"Extracting Neutron Star Radii from the Tidal Deformabilities of GW170817." Talk at the Institute for Nuclear Theory's First Multi-Messenger Observations of a Neutron Star Merger Workshop in Seattle. March 2018.

"Confronting Models of Massive Star Evolution and Explosions with Remnant Mass Measurements." Poster at the International Astronomy Union Symposium "50 Years of Pulsars," at Jodrell Bank Observatory in the U.K., September 2017.

"Neutron Star Observables to the Equation of State." Talk at the TCAN Annual Collaboration Meeting in Cambridge, MA. October 2016.

"From Neutron Star Observables to the Dense Matter Equation of State." Talk at the Institute for Nuclear Theory's Phases of Dense Matter Workshop in Seattle. July 2016.

"Constraining Neutron Star EOS from Moment of Inertia Measurements." Talk at the TCAN Annual Collaboration Meeting in Tucson. December 2015.

"Radio Pulsar Emission and Birth Studies." Talk to the American Physical Society Leroy Apker Award Selection Committee in Chicago. September 2015.

"Characterizing the Unusual Emission of PSR J0901-4624." Talk at the Carleton College Department of Physics and Astronomy. October 2014.

"Characterizing the Unusual Emission of PSR J0901-4624." Poster Presentation at the Carleton College and St. Olaf College Physics Research Symposium. October 2014.

"Jitter in the PPTA Pulsars." Talk at Parkes Pulsar Timing Array (PPTA) meeting, at the CSIRO Australia Telescope National Facility in Sydney, Australia. August 2013.

LEADERSHIP AND SERVICE

- Graduate Student Representative on the Steward Observatory Committee for Diversity and Inclusivity, May 2017 – Present.
- Graduate Student Representative on the Graduate Admissions Committee for Steward Observatory, Winter 2017/2018.
- Volunteer judge for Astronomy, Physics, and Math projects at the Southern Arizona Research, Science, and Engineering Foundation (SARSEF) Science Fair, March 2017, 2018, and 2019.
- Physics Student Departmental Advisor, Carleton College, 2014-2015.
- Carleton College Women in Physics Society, Co-President, 2014-2015.